

DC: Reinforcement Learning for Context-aware and Adaptive Lighting Design using Extended Reality: Impacts on Human Emotions and Behaviors

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ABSTRACT

In the interconnected world of smart built environments, Extended Reality (XR) emerges as a transformative technology that can enhance user experiences through personalized lighting systems. The integration of XR with deep reinforcement learning for adaptive lighting design (ALD) has potential to optimize visual experiences while addressing real-time data analysis, XR system complexities, and integration challenges with building automation systems. The research centers on developing an XR-based ALD system that dynamically responds to user preferences, thereby positively impacting human emotions, behaviors, and overall well-being.

Index Terms: Context-Aware Lighting Design—Deep Reinforcement Learning Algorithms—Extended Reality—Smart Built Environment, Adaptive Lighting Design, Human-Computer Interaction

1 BACKGROUND AND MOTIVATION

The advent of Extended Reality (XR) and its integration with advanced computational models like Deep Reinforcement Learning (DRL) and Large Language Models (LLMs) like ChatGPT has opened new frontiers in smart built environment (SBE) design, particularly in the domain of adaptive lighting design (ALD). This research is motivated by the potential of these technologies to revolutionize lighting design, transcending the limitations of traditional systems and fostering environments that dynamically respond to human needs and behaviors.

Recent advancements in XR technologies, underscore the growing relevance of immersive simulations in understanding and optimizing human-centric environments. These studies provide a foundation for exploring how XR can be effectively used for lighting design simulations in SBEs, offering insights into user-centered design and human factors in lighting. The systematic review by Mohammadrezaei et al. in particular, emphasizes the importance of extended reality in the systematic exploration of lighting designs and their impacts [1, 2].

Similarly, the emergence of LLMs, has shown how these models can serve as decision-makers in complex scenarios, such as autonomous driving. This opens up possibilities for employing LLMs like ChatGPT in smart lighting systems, where they can interact with users to understand and adapt to their preferences in real-time. Using LLMs promises a more intuitive and user-friendly approach to lighting control, moving beyond pre-set configurations to a more fluid and responsive system [5].

The potential for energy savings and optimization of lighting systems, also forms a critical part of this research's motivation. Their focus on maximizing daylight utilization while minimizing occupant impact aligns with the goal of creating adaptive lighting systems that are not only user-friendly but also energy-efficient [3].

For example, Rajalakshmi et al. demonstrate the application of adaptive fuzzy logic in deep-learning models for underwater

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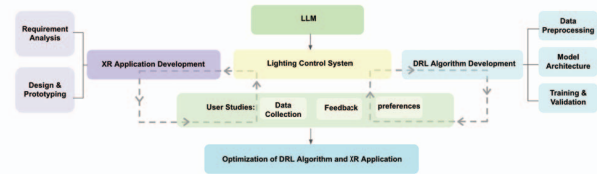


Figure 1: ALD framework.

communication systems, indicating the versatility and efficacy of adaptive models in various settings. This inspires the exploration of similar ALD models, particularly in the context of SBEs where user comfort and system efficiency are paramount [4].

The proposed research builds upon these foundational studies to develop a framework for lighting system design that integrates XR, DRL and LLMs (Figure 1). The framework does more than just cater to aesthetic and functional needs; it aims to intuitively align with users' emotional and behavioral patterns, thereby enriching the overall experience in SBEs.

We aim to address the limitations of traditional lighting systems by developing an adaptive, context-aware solution that resonates with users' emotional and behavioral patterns. This research is driven by the desire to enhance user experiences in SBEs, leveraging the latest advancements in XR and Artificial Intelligence (AI) to create lighting systems that are not only responsive to user preferences but also contribute to energy efficiency and sustainability.

2 PROPOSED RESEARCH

The proposed research unfolds into three phases, each addressing a core aspect of integrating DRL with XR for ALD.

The journey begins with the development of an XR application specifically for ALD in SBEs. This phase focuses on comprehensive requirement analysis, engaging with lighting designers, XR experts, and potential users to pinpoint critical functional and experiential needs. Following this, the design and prototyping stage is critical in shaping a user interface that provides ease of use and a realistic simulation of varied lighting conditions. A significant innovation in this phase is the integration of LLMs into the application. These LLMs are designed to interact with users, querying their lighting preferences in a conversational, intuitive manner. This approach replaces traditional sensor-based data collection, offering a more user-centric and interactive method of understanding individual lighting preferences. Preliminary testing ensures the application's stability and its ability to accurately simulate a wide range of lighting scenarios.

The second phase encompasses comprehensive user studies, designed to collect detailed data on user interactions with the ALD system within the XR environment. A diverse group of participants, representing various demographics and preferences, will ensure a broad understanding of different user needs. The study is structured to observe and record interactions in controlled XR settings, focusing on user satisfaction, ease of use, and lighting preferences. A blend

of qualitative and quantitative research instruments will be used, including surveys, interviews, and sensors, to collect and analyze data ranging from subjective feedback to psycho-physiological responses. This will allow identifying user behavior patterns, preferences, and areas for enhancement in the ALD system.

The final phase is centered on the development and refinement of a DRL algorithm for personalized lighting control. This phase involves the intricate process of data preprocessing, where data from user studies is cleaned and structured for training the DRL model. Developing the model is an iterative process, aimed at accurately predicting optimal lighting conditions based on user inputs and environmental factors. The training and validation of the model are crucial, with various performance metrics applied to refine its accuracy and responsiveness. A separate dataset is used for validation to ensure the model's effectiveness in real-world applications. The culmination of this phase is the integration of the optimized DRL model with the XR application, followed by comprehensive testing to evaluate the system's overall performance in providing dynamic, personalized lighting control.

3 RESEARCH GOALS AND EXPECTED CONTRIBUTIONS

The main research focus is investigation of the transformative potential of personalized lighting design and control for SBEs as the traditional tools employed in lighting design are inadequate for creating a design that considers the subjective responses of the user, as they lack the capability to generate an immersive virtual environment. Specifically, the study aims to explore the influence of DRL algorithms on shaping the emotional experiences, behavioral patterns, and overall well-being of individuals immersed in these technologically advanced environments. The research goals include:

RG1: Development of an XR Application with Integrated LLMs. Develop an ALD XR application that integrates LLMs for an interactive, user-centric approach. The LLMs will engage users in natural language to understand and adapt to their lighting preferences.

RG2: Assessing User Interaction and Satisfaction. Conduct comprehensive user studies to evaluate user interactions with the XR application. Use the LLMs within the application to gather qualitative and quantitative feedback on user preferences, comfort levels, and the overall user experience.

RG3: Development and Optimization of a DRL Algorithm for Personalized Lighting. Develop and refine a DRL algorithm that uses data from user interactions (collected via LLMs) to dynamically adjust lighting.

Expected Contributions: ALD pioneers the use of XR and DRL for dynamic, user-responsive environments, enhancing personal comfort and well-being in SBEs. The innovative integration of LLMs for user interaction in XR settings represents a significant advancement in human-computer interaction, offering intuitive and natural ways of gathering user preferences. Methodologically, this research contributes to AI by developing and refining a DRL algorithm based on user data, showcasing a novel application of DRL in real-world scenarios. Collectively, these contributions are anticipated to offer valuable insights and set new benchmarks in SBE control, user-centric design, and the intersection of AI with environmental psychology.

The research questions guiding this study are as follows:

RQ1: How to integrate XR and LLMs to effectively address diverse user preferences for lighting aesthetics and intensity?

RQ2: What is the impact of LLM-driven user interaction on the adaptability and performance of the DRL algorithm for personalized lighting control?

RQ3: How to customize DRL algorithms for different user profiles based on their interactions with LLMs in XR environments?

RQ4: How do users' emotional states and behavior correspond to changes in personalized lighting conditions, and how does ADL impact user behavior and emotions in the same context?

To effectively address our research questions, we undertake a comprehensive and integrated approach centered around the development and analysis of an XR application augmented with ChatGPT as the core large language model. ChatGPT's advanced conversational capabilities will play a crucial role in interactively determining user preferences for lighting aesthetics and intensity, thereby ensuring a highly personalized lighting experience in XR environments. The data gathered from user interactions with ChatGPT will help enhance the adaptability and performance of our DRL algorithm. This approach is designed to enable the algorithm to learn and adjust to user preferences dynamically and in real-time. Furthermore, by leveraging the detailed insights obtained from ChatGPT-user dialogues, the DRL algorithm will be finely tuned to accommodate the diverse lighting preferences and sensitivities of individual user profiles. Additionally, the research will include exploration of the emotional and behavioral impacts on users responding to the personalized lighting adjustments suggested by the DRL algorithm, as informed by ChatGPT interactions. This holistic strategy, combining the capabilities of ChatGPT with innovative XR and AI technologies, is also expected to provide valuable contributions to SBE design and offer novel insights into the enhancement of user experiences in intelligent, adaptive spaces.

4 CONCLUSION

This research investigates the integration of Deep Reinforcement Learning with Extended Reality and ChatGPT for Adaptive Lighting Design in Smart Built Environments. Focusing on overcoming the constraints of traditional lighting systems, the goal is to develop an ALD framework for dynamic and responsive XR applications. ChatGPT's conversational intelligence plays a key role in capturing and interpreting user preferences, enabling the DRL algorithm to tailor lighting settings in real-time. Through this innovative approach, the research aims not only to enhance user experiences but also to provide insights into the psycho-physiological effects of personalized lighting, marking a significant step forward in the harmonization of technology and human-centric design in SBEs.

REFERENCES

- [1] E. Mohammadrezaei, S. Ghasemi, P. Dongre, D. Gračanin, and H. Zhang. Systematic review of extended reality for smart built environments lighting design simulations. *IEEE Access*, pp. 1–1, 2024. doi: 10.1109/ACCESS.2024.3359167
- [2] E. Mohammadrezaei, A. Giovannelli, L. Lane, and D. Gračanin. A digital twin based approach to smart lighting design. In *2022 Winter Simulation Conference*, 13 Dec. 2022.
- [3] M. Papinutto, R. Boggetti, M. Colombo, C. Basurto, K. Reutter, D. Lalanne, J. H. Kämpf, and J. Nembrini. Saving energy by maximising daylight and minimising the impact on occupants: An automatic lighting system approach. *Energy and Buildings*, 268:112176, 2022.
- [4] R. Rajalakshmi, S. Pothiraj, M. Mahdal, and M. Elangovan. Adaptive fuzzy logic deep-learning equalizer for mitigating linear and nonlinear distortions in underwater visible light communication systems. *Sensors*, 23(12):5418, 2023.
- [5] H. Sha, Y. Mu, Y. Jiang, L. Chen, C. Xu, P. Luo, S. E. Li, M. Tomizuka, W. Zhan, and M. Ding. Languagempc: Large language models as decision makers for autonomous driving. *arXiv preprint arXiv:2310.03026*, 2023.